TT-L-132
March 28, 1966
SUPERSEDING
Fed. Spec. TT-W-132
April 23, 1951

FEDERAL SPECIFICATION

LITHOGRAPHIC PLATE ASPHALTUM SOLUTION

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 This specification covers one type and grade of lithographic plate asphaltum solution (see 6.1).

2. APPLICABLE DOCUMENTS

2.1 Specifications and standards. The following specifications and standards, of the issues in effect on date of invitation for bids or request for proposal, form a part of the specification to the extent specified herein:

Federal Specification:

PPP-C-96—Cans, Metal, 28 Gage and Lighter.

Federal Standards:

Fed. Std. No. 123—Marking for Domestic Shipment (Civilian Agencies).

Fed. Test Method Std. No. 141—Paint, Varnish, Lacquer, and Related Materials; Methods of Inspection, Sampling, and Testing.

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U. S. Government Printing Office, Washington, D. C. 20402.

(Single copies of this specification and other product specifications required by activities outside the Federal Government for bidding purposes are available without charge at the General Services Administration Regional Offices in Boston, New York, Washington, D. C., Atlanta, Chicago, Kansas City, Mo., Dallas, Denver, San Francisco, Los Angeles, and Seattle, Wash.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

Military Standards:

MIL-STD-105—Sampling Procedures and Tables for Inspection by Attributes.

MIL-STD-129—Marking for Shipment and Storage.

(Copies of Military Specifications and Standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publication. The following document forms a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

U. S. Department of Health, Education, and Welfare, Food and Drug Administration

Federal Hazardous Substances Labeling Act.

(Application for copies should be addressed to the U. S. Department of Health, Education, and Welfare, Food and Drug Administration, Washington, D. C., 20315.)

3. REQUIREMENTS

3.1 Material. The asphaltum solution shall be a solution of asphaltum in turpentine or volatile mineral spirits.

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3.1.1 *Nonvolatile matter (at 105° ± 2° C.).* The asphaltum solution shall contain not less than 18 percent, by weight, of nonvolatile matter when tested as specified in 4.3.6.

3.1.2 *Water.* The asphaltum solution shall contain not more than 0.5 percent, by weight, of water when tested as specified in 4.3.6.

3.1.3 *Grit.* The asphaltum solution shall be free of grit and shall not scratch microscope slides when tested as specified in 4.3.6.

3.2 **Performance.** When tested as specified in 4.3.6, the asphaltum solution shall protect the plate so that a press test shall show that the image area is receptive to ink and that the nonprinting areas of the test plate are clean and do not accept printing ink.

3.3 **Workmanship.** The asphaltum solution shall be a clean, homogenous liquid, free of sediment and suspended matter, and shall conform to the quality levels established herein.

4. QUALITY ASSURANCE PROVISIONS

4.1 **Responsibility for inspection.** Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 **Inspection.** Sampling for inspection shall be in accordance with MIL-STD-105, except where otherwise indicated.

4.2.1 *Component and material inspection.* In accordance with 4.1, components and materials shall be inspected and tested in accordance with all the requirements of referenced specifications, drawings, and standards unless otherwise excluded, amended, modified, or qualified in this specification or applicable purchase documents.

4.3 **Inspection of the end item.**

4.3.1 *Examination of the end item.* The end item and its preparation for delivery shall be examined for the defects in the applicable subparagraphs at the inspection levels and acceptable quality levels (AQLs) set forth in 4.3.5. The lot size shall be expressed in unit containers of the specified capacity for the examinations in 4.3.2 and 4.3.3 and in units of shipping containers for the examination in 4.3.4.

4.3.2 *Examination of contents of unit containers for defects in appearance and condition.* The sample unit shall be one filled unit container.

Examine	Defect
Workmanship	Not uniform liquid.
	Contains sediment, dirt, or insoluble matter.
	Separates into liquid or solid phase.

4.3.3 *Examination of the end item for defects in net contents.* The sample unit for this examination shall be one filled unit container. The lot shall be unacceptable if the average net content per container examined is less than the quantity specified.

4.3.4 *Examination of preparation for delivery.* An examination shall be made to determine that packaging, packing, and markings comply with the requirements of section 5. Defects shall be as scored below. The sample unit shall be one shipping container, fully packed, selected just prior to the closing operation. Closure defects listed below shall be examined on closed shipping containers fully prepared for delivery.

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Examine	Defect
Packaging and packing	Not in accordance with contract requirements.
	Container not as specified; closures not accomplished by specified or required methods or materials.
	Any nonconforming component; component missing, damaged, or otherwise defective.
	Inadequate application of components such as: incomplete closure of case liners, container flaps, loose or inadequate sealing, strapping or stapling.
	Bulged or distorted containers.
Markings (exterior and interior, as applicable)	Incorrect; incomplete; illegible; omitted; of improper size, location, sequence, method of application, or not in accordance with contract requirements.
	Not in accordance with the Federal Hazardous Substances Labeling Act.
	Legend not as specified (see 5.3.1.1).
	Directions for use missing.

4.3.5 *Inspection levels and AQLs for examinations.* The inspection levels, for determining the sample size, and the AQLs, expressed in defects per 100 units, shall be as follows:

Examination paragraph	Inspection level	AQLs
4.3.2	I	4.0
4.3.3	S-3	—
4.3.4	S-1	2.5

4.3.6 *Testing of the asphaltum solution.* The asphaltum solution shall be tested for the applicable characteristics as specified in table I for each inspection lot. The sample unit shall be a one-pint sample. The sample size shall be S-1, except that at least five sample units shall be randomly selected throughout the lot. A one-quart composite shall be prepared by taking an equal volume from each sample unit. The composite sample shall be placed in a clean, dry bottle and sealed. Care shall be exercised in preparing the composite sample and during testing (as well as during sampling and storage), to prevent loss of volatile material and product degradation as well as contamination. The lot shall be unacceptable if the composite sample fails to meet any test requirement specified.

4.4 **Test procedures.**

4.4.1 *Nonvolatile content (at 105° ± 2° C.).* Determine the nonvolatile content of the asphaltum solution in accordance with method 4041 of Fed. Test Method Std. No. 141, using approximately a 5-gram sample.

4.4.2 *Water.* Determine the amount of water present in 100 grams of the asphaltum solution in accordance with method 4081 of Fed. Test Method Std. No. 141.

4.4.3 *Grit.* Spread a thin film of asphaltum solution, taken from a thoroughly agitated container, over a clean microscope slide. Place another slide over the film and press the slides together lightly. Rub one slide over the other with a rotary motion. The movement between the slides shall be

TABLE I. *Instructions for testing—asphaltum solution*

Characteristic	Specification reference		Requirements applicable to lot aver.	Number determinations per composite sample	Results reported as[1]	
	Requirement	Test method			Pass or fail	Numerically to nearest
Nonvolatile content	3.1.1	4.4.1	X	2		Average of 2 determinations to 0.5 percent.
Water	3.1.2	4.4.2	X	2		Average of 2 determinations to 0.01 percent.
Grit	3.1.3	4.4.3	X	1	X	
Performance test	3.2	4.4.4	X	1	X	

[1] If failure is indicated, report either description of failure or the numerical point of failure, as applicable.

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smooth with no indication of grit. The slides when wiped clean of the asphaltum solution shall not be scratched.

4.4.4 *Performance test.*

4.4.4.1 *Preparation of test plate.* Prepare a fresh (manufactured not less than 3 months prior to use) aluminum, presensitized plate by exposing and developing it in accordance with manufacturer's directions.

4.4.4.2 *Procedure.* Place a test plate, 20 by 22-5/8 inches with medium work coverage and which has been properly gummed with an 8° Baume gum arabic solution, on a dry table. Pour one ounce of the asphaltum solution onto the gummed surface of the test plate and spread uniformly with a small wad of cotton. Remove the excess with a fresh wad of cotton and polish until dry. After 48 hours, flood the coated plate with water and rub lightly with a wad of cotton. The asphaltum coating shall be readily removed. Place the plate on the press and make a short run to ascertain that the image area is receptive to ink and that the asphaltum solution did not remain on the nonprinting area of the plate.

5. PREPARATION FOR DELIVERY

5.1 Packaging. Packaging shall be level A or C as specified (see 6.2).

5.1.1 *Level A.* The asphaltum solution shall be supplied in one-quart or one-gallon cans conforming to type V, class 4 of PPP-C-96, as specified (see 6.2 and 6.3).

5.1.2 *Level C.* The asphaltum solution shall be packaged to afford adequate protection against damage during shipment from the supply source to the first receiving activity. The supplier may use his standard practice when it meets this requirement.

5.2 Packing. Packing shall be level A, B, or C as specified (see 6.2).

5.2.1 *Level A.* The asphaltum solution, packaged as specified in 5.1, shall be packed in accordance with the level A requirements in the appendix of PPP-C-96.

5.2.2 *Level B.* The asphaltum solution, packaged as specified in 5.1, shall be packed in accordance with the level B requirements in the appendix of PPP-C-96.

5.2.3 *Level C.* The asphaltum solution, packaged as specified in 5.1, shall be packed in a manner to insure carrier acceptance and safe delivery at destination at the lowest transportation rate for such supplies. Containers shall be in accordance with rules or regulations of carriers applicable to the mode of transportation.

5.3 Marking.

5.3.1 *Civil agencies.* In addition to any special marking required by the contract or order, shipping containers shall be marked in accordance with Fed. Std. No. 123.

5.3.1.1 *Labeling.* Labeling shall conform to the regulations under the Federal Hazardous Substances Labeling Act, Part 191. In addition, each can shall be marked with the name of the material, the quantities contained therein, the name and address (with or without) the trademark of the supplier, and the following in 18-point type:

"DANGER: CONTENTS VOLATILE AND COMBUSTIBLE.
AVOID INHALATION OF FUMES.
PROVIDE ADEQUATE VENTILATION IN WORK AREAS.
HARMFUL OR FATAL IF SWALLOWED.
AVOID PROLONGED OR REPEATED CONTACT WITH SKIN."

Directions for use shall also appear on each can.

5.3.2 *Military requirements.* In addition to any special marking required by the con-

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tract or order, shipping containers shall be marked in accordance with MIL-STD-129.

5.3.2.1 *Labeling.* In addition to the interior package marking requirements of MIL-STD-129, each can shall be marked as specified in 5.3.1.1 herein.

6. NOTES

6.1 **Intended use.** The asphaltum solution covered by this specification is intended for use as a protective coating for images on offset duplicating plates, and to make the image areas more ink receptive. The material is to be applied after developing ink has been removed with a solvent.

6.2 **Ordering data.** Purchasers should select the preferred options permitted herein and include the following information in procurement documents:

(*a*) Title, number, and date of this specification.
(*b*) Selection of the applicable levels of packaging and packing (see 5.1 and 5.2).
(*c*) If asphaltum is to be supplied in one-quart or one-gallon cans (see 5.1.1).

6.3 **Packaging.** The method of packaging referred to in 5.1.1 is similar to method III of MIL-P-116.

MILITARY CUSTODIANS:

Army—GL
Navy—MC
Air Force—68

Review activities:

Army—GL, MD, MU
Air Force—68

User activities:

Army—MR, MO
Navy—MC

Preparing activity:

Army—GL

Review and user information is current as of the date of this document. For future coordination of changes to this document, draft circulation should be based on the information in the current Federal Supply Classification Listing of DOD Standardization Documents.

Orders for this publication are to be placed with General Services Administration, acting as an agent for the Superintendent of Documents. See section 2 of this specification to obtain extra copies and other documents referenced herein. Price 5 cents each.

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SPECIFICATION ANALYSIS SHEET

Form Approved Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity.

SPECIFICATION: Lithographic Plate Asphaltum Solution - TT-L-132

ORGANIZATION	CITY AND STATE

CONTRACT NO.	QUANTITY OF ITEMS PROCURED	DOLLAR AMOUNT
		$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)*	DATE

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